

Latency-Cost Frontier

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Speed you pay for vs. speed you need

PROFESSIONAL DEFINITION

The Latency-Cost Frontier expresses the trade-off between the speed of an AI system and its cost, showing that lower latency generally commands higher cost and that not all use cases need the fastest tier.

WHO DEVELOPED IT

An engineering-economics construct from systems and ML-serving practice.

WHY IT WAS DEVELOPED

It was developed to prevent over-spending on speed that the use case does not require, matching latency to genuine need.

FIGURE 1 · LATENCY-COST FRONTIER

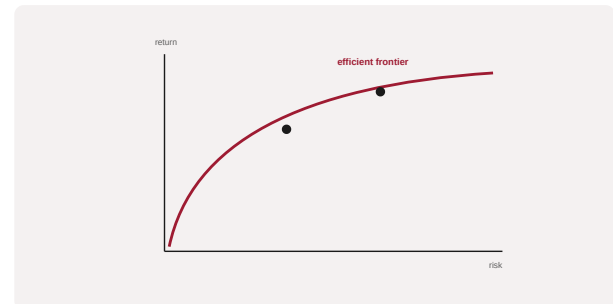


Figure 1. A schematic of the Latency-Cost Frontier as applied in BARBM312 (illustrative).

HOW IT IS USED

Use cases are placed on the frontier; each is served at the cheapest tier that meets its real latency requirement.

WHEN IT IS USED

When designing AI serving architecture and controlling inference cost.

BY WHOM IT IS USED

ML engineering, product and finance teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the speed-versus-cost trade-off explicit per use case, avoiding paying for latency that delivers no value.

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Cloud Cost Waterfall · Slibrary 10

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SOURCES (HARVARD REFERENCING STYLE)

Dean, J. and Barroso, L.A. (2013) 'The tail at scale', Communications of the ACM, 56(2).

Crankshaw, D. et al. (2017) 'Clipper: A low-latency online prediction serving system', NSDI.